Jamie Seoh

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Education

University of Waterloo Sep 2025 – May 2030
Bachelor of Software Engineering *3.95/4.00 (91.2%) GPA*

- Top 7.6% in Euclid Math Contest, 2 x Hackathon Winner, Software Engineering Entrance Scholarship

Experience

Cerebras Systems May 2026 – Aug 2026
AI Software Engineer Intern *Toronto, ON*

- Engineered a **fault-injection** tool with C++ and assembly spanning runtime controls and on-device kernels to simulate crashes, stalls, and slowdowns, reducing average customer-failure triage time by **2 days**.
- Refined fault detection to separate likely faulty devices from affected neighbors, reducing major reports flagged for on-call escalation from **11 to 1** in one production-stall replay.
- Built a breakpoint managing **concurrent requests** and exposing counters to debug a fatal KV-cache refill bug.
- Added deterministic GTest & pytest regression tests to catch unintended worker pauses and verify delay timing.
- Developed a worktree and agent management dashboard to streamline parallel development for fellow interns.

University of Waterloo Sep 2026 – Present
Undergraduate Research Assistant *Waterloo, ON*

- Researching **Rust-to-LLVM** IR translation and verification semantics under Prof. Arie Gurfinkel to extend Rust support for SeaHorn, a tool for automated program verification.
- Wrote verification harnesses in Rust for **pointer aliasing** and memory updates; repaired unreachable and compiler-eliminated assertions, validated through vacuity and mutation checks.

UW Association of Korean-Canadian Scientists Aug 2026 – Present
Full-Stack Developer *Waterloo, ON*

- Leading a Vite website rebuild for **100+ members**, redesigning member-facing pages and CMS admin workflows.
- Consolidated storage and APIs on **Cloudflare**, simplifying infrastructure management by migrating from AWS.
- Designed transactional safeguards against concurrent admin lockout, with audit logs tracking permission changes.
- Built **CI/CD** pipelines for Cloudflare staging, gating deployment on application, migration, and permission tests.

Projects

Quota — DeltaHacks12 First Place | *TypeScript, LangChain, MongoDB*

- Built a **TypeScript** VS Code extension for inline cost annotations, heatmaps, and one-click optimizations.
- Indexed **10,000+**-line codebases in under **3 seconds** using **AST**-based inefficiency detection.
- Cut developer API spend by up to **40%** by providing real-time API cost insights via regex and usage caching.
- Created a React Flow sandbox for budget-aware system planning with **RAG Gemini** chatbot.

Nebula AI | *React Native, FastAPI, Docker, GCP*

- Shipped a **React Native** app turning handwritten notes into searchable content with LaTeX and AI summaries.
- Enabled semantic search to fetch notes by meaning, reducing retrieval latency by **40%** with **IVFFlat indexing**.
- Refined note uploads based on user feedback, adding image previews, processing status, and retry controls.

Community

Hack the North Feb 2026 – Present
Logistics Organizer *Waterloo, ON*

- Led end-to-end operations for **300+ volunteers** supporting Hack the North, a **1,000+**-participant hackathon.
- Redesigned the volunteer shift-selection system to prevent assignment race conditions and streamline signup.

Technical Skills

Languages: Python, C/C++, Bash, TypeScript & JavaScript, HTML & CSS, SQL, ASM, Scala
Frameworks and Libraries: PyTorch, NumPy, GTest, pytest, React, React Native, Next.js, FastAPI, LLVM
Databases and APIs: MongoDB, PostgreSQL, GraphQL, REST APIs, Prisma
Technologies: Docker, Git/GitHub, Linux, GCP, AWS (Lambda, S3), Cloudflare, Azure, Figma